

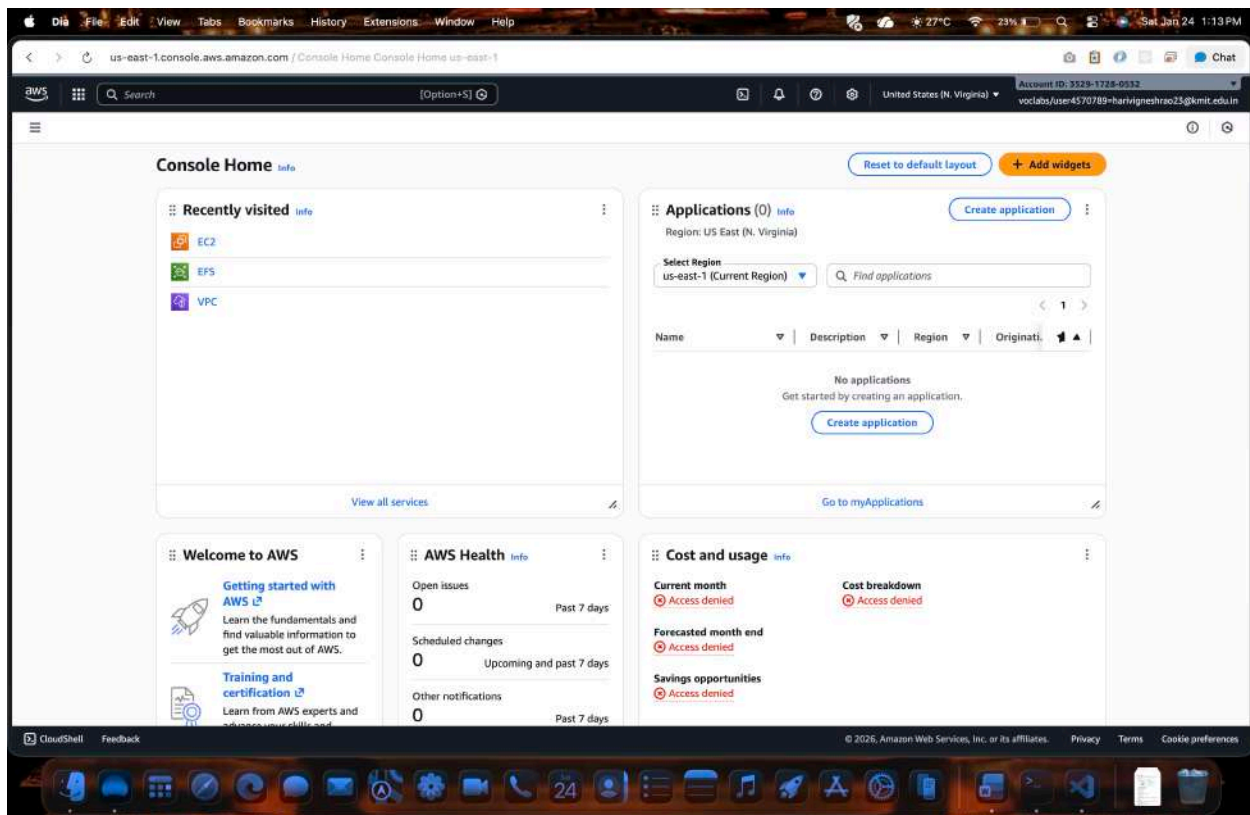
Launch an EC2 Instance and Connect

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Procedure

Step 1: Login to AWS / AWS Academy Learner Lab

Step 2: Choose region which is near ? (Asia pacific – Mumbai/N.Virigina)



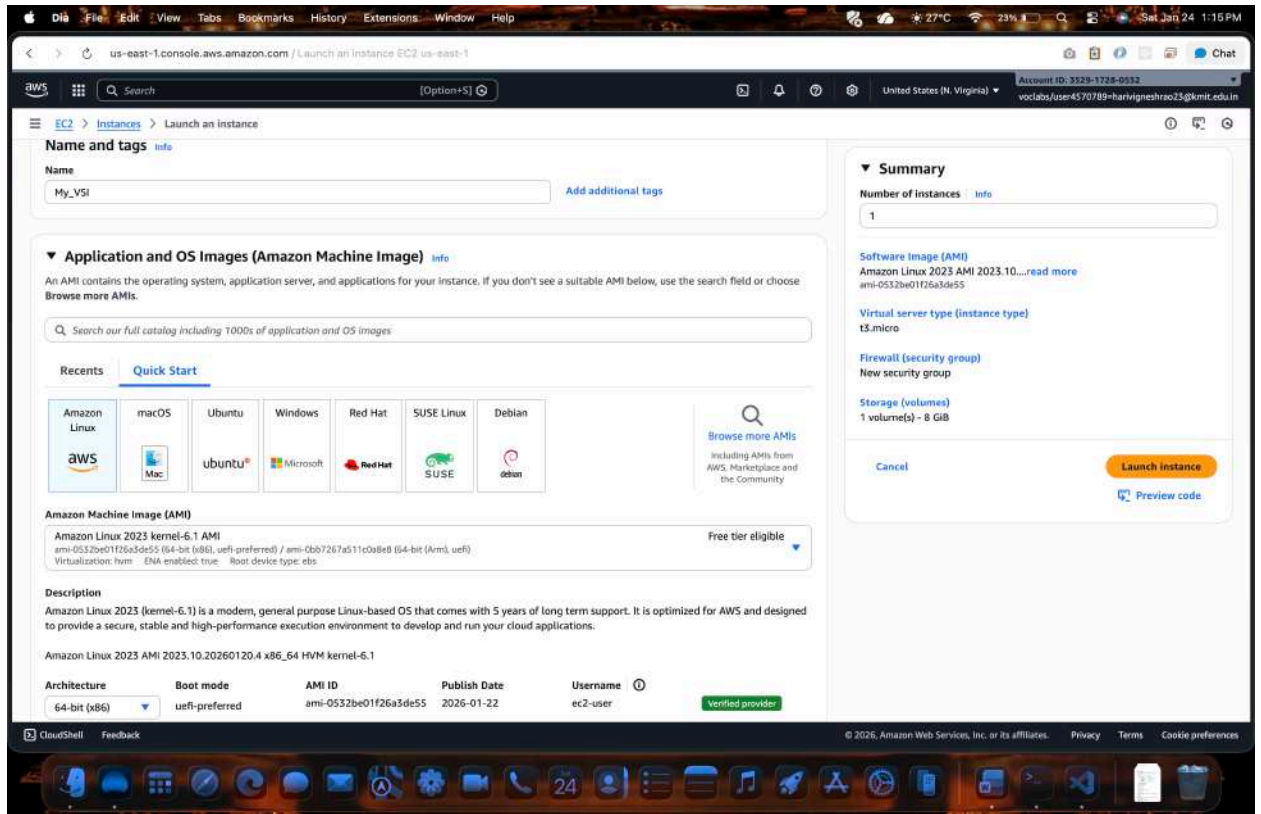
Step 3: Services -- EC2

Services -- EC2 --- Launch Instance

Step 1 --Name (Giving name to the machine) My_VS1

Step 2 -- Choose AMI (Amazon Machine Image – e.g., Amazon Linux, Ubuntu)

Step 3 -- Select Architecture [default 64-bit(x86)]

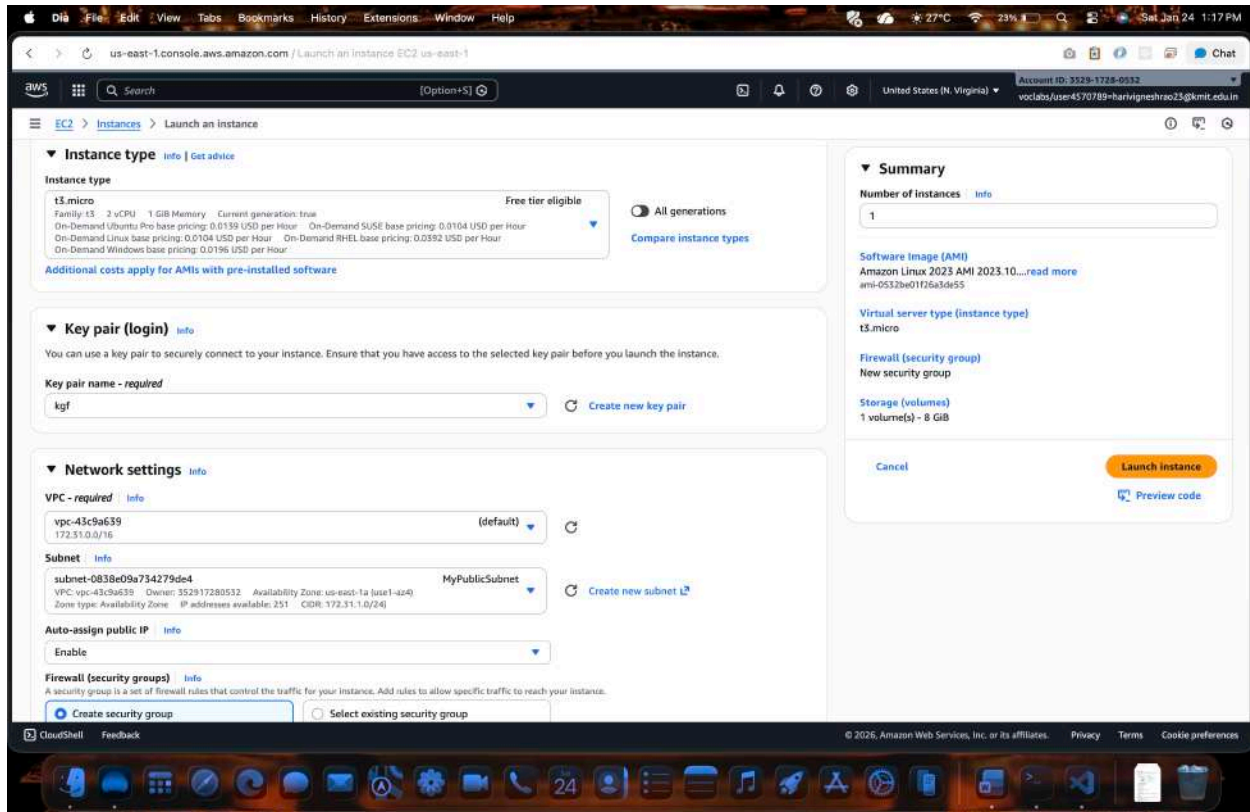


Step 4 -- Select Instance Type (default t3.micro, [2 CPU, 1GB Memory])

Step 5 --Create a new keypair [type-RSA – file format .pem]

Step 6 -- Configure network, storage, and security group (Default values are ok)

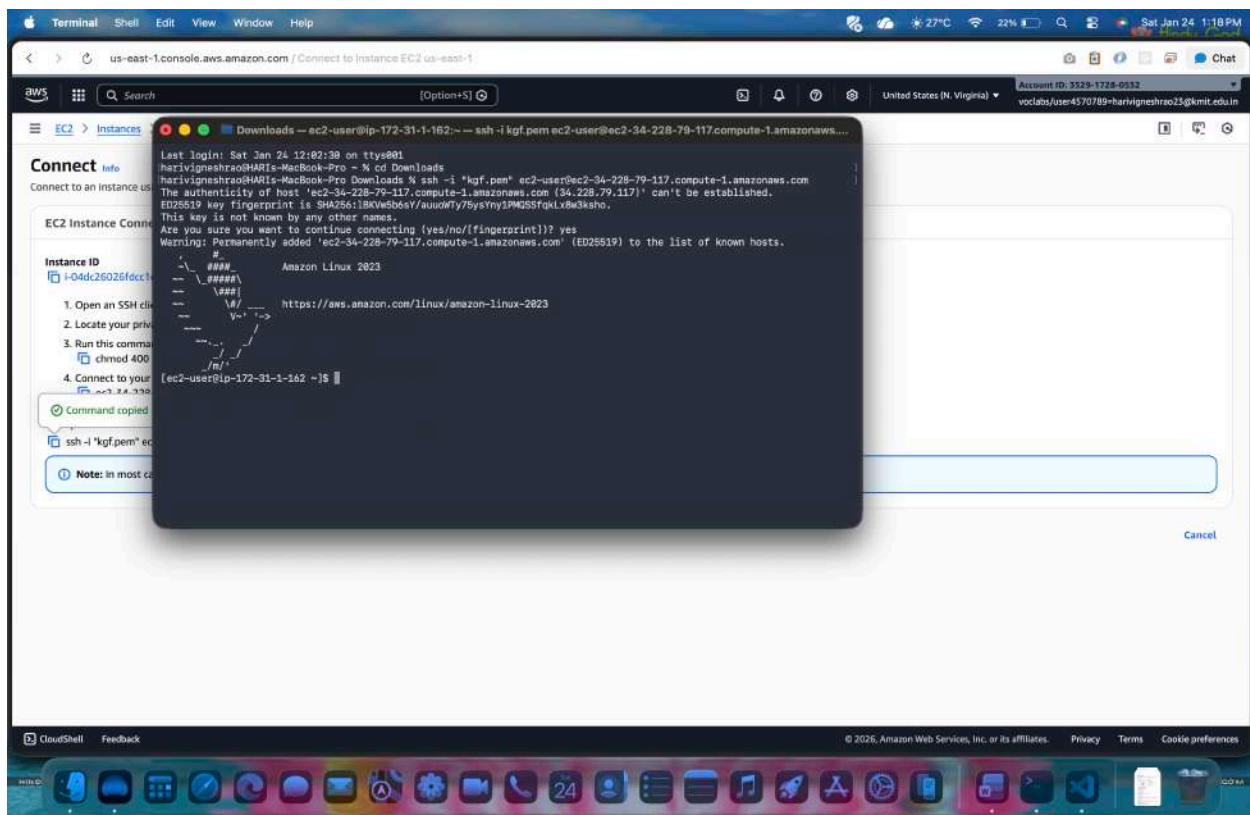
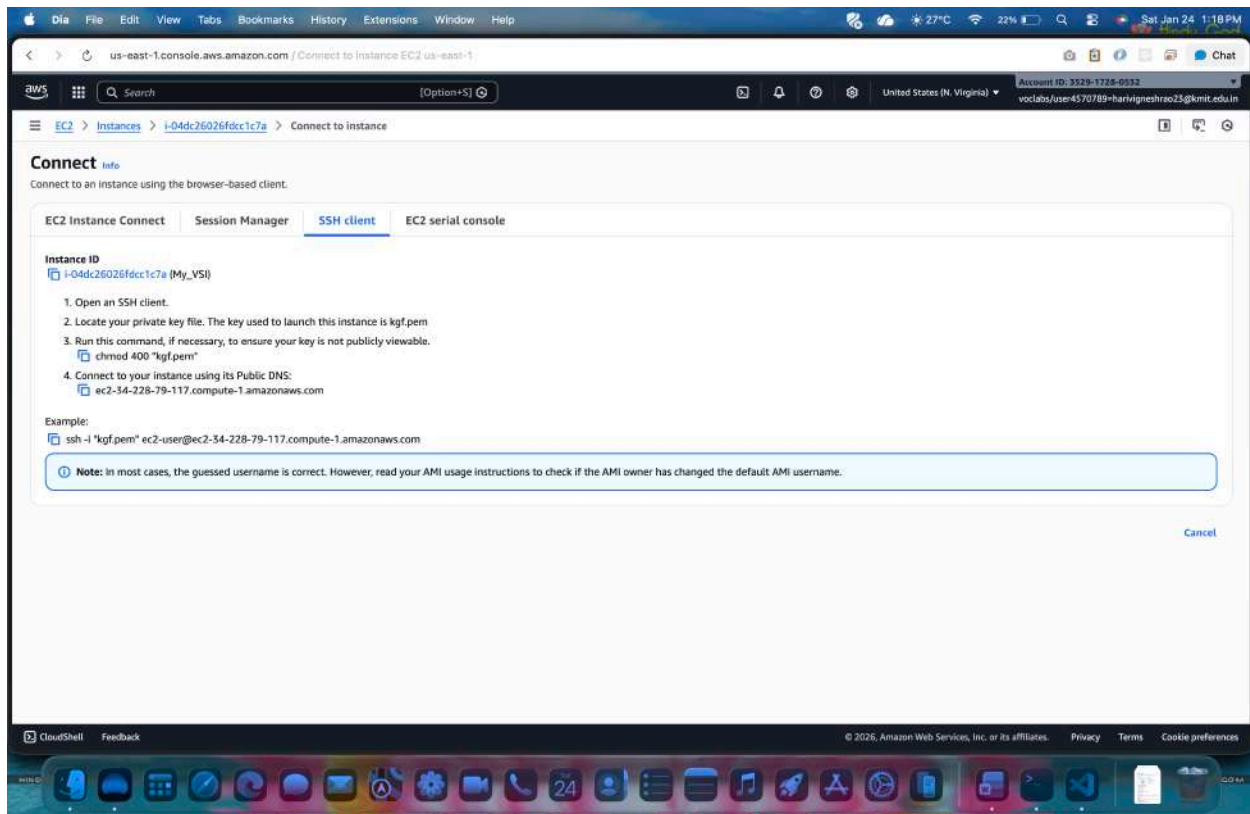
Stage 7 --- click on launch instance.



Connect EC2 Ubuntu using SSH:

Steps:

1. Open the terminal (powershell/gitbash on windows)
2. Change the directory to the key location
3. Connect using SSH [for ubuntu : `ssh -i mykey.pem ubuntu@<Public-IP>`]
4. Successfully connected. The ubuntu command interface is active.



Connect EC2 Ubuntu Using PuTTY (.ppk Key)

◆ STEP 1: Check Security Group

Go to **EC2** → **Security Groups** → **Inbound rules**

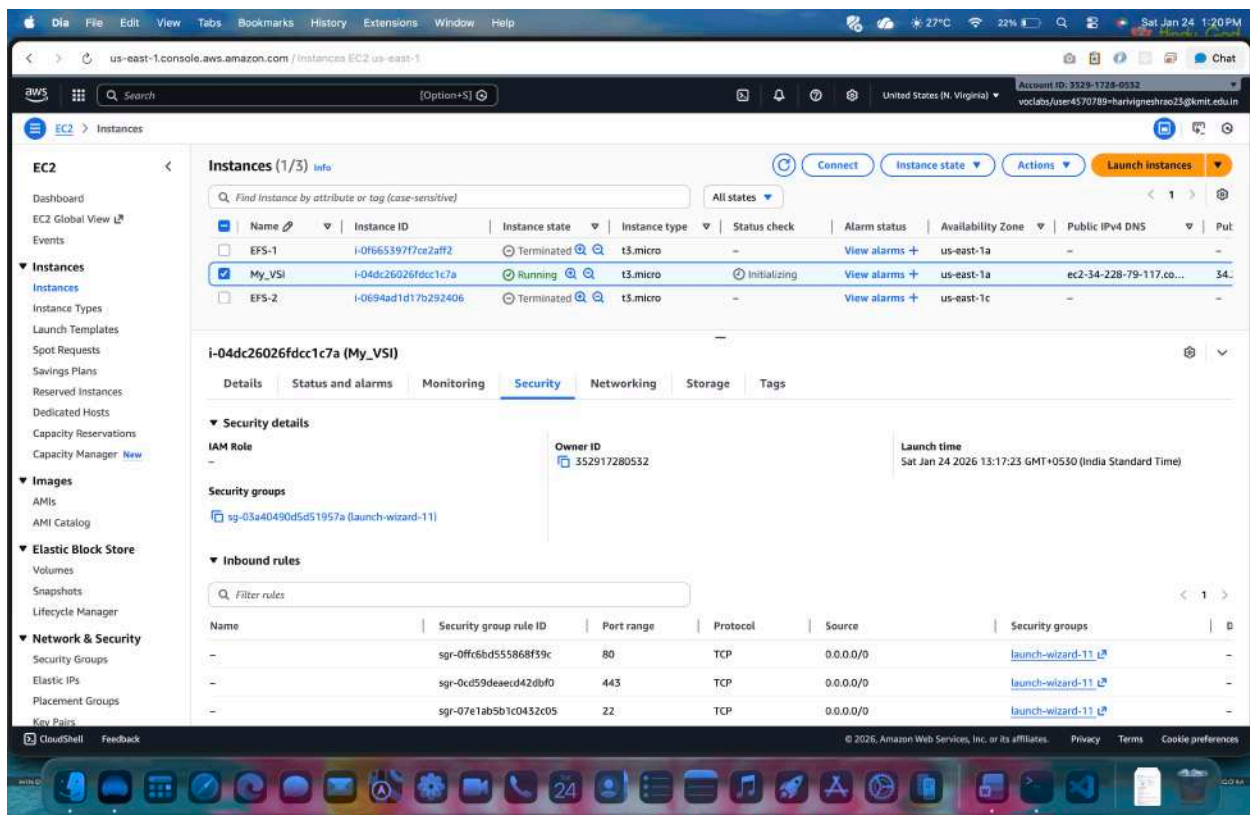
Ensure this rule exists:

Type: SSH

Port: 22

Source: My IP

✗ If SSH is blocked → connection will fail

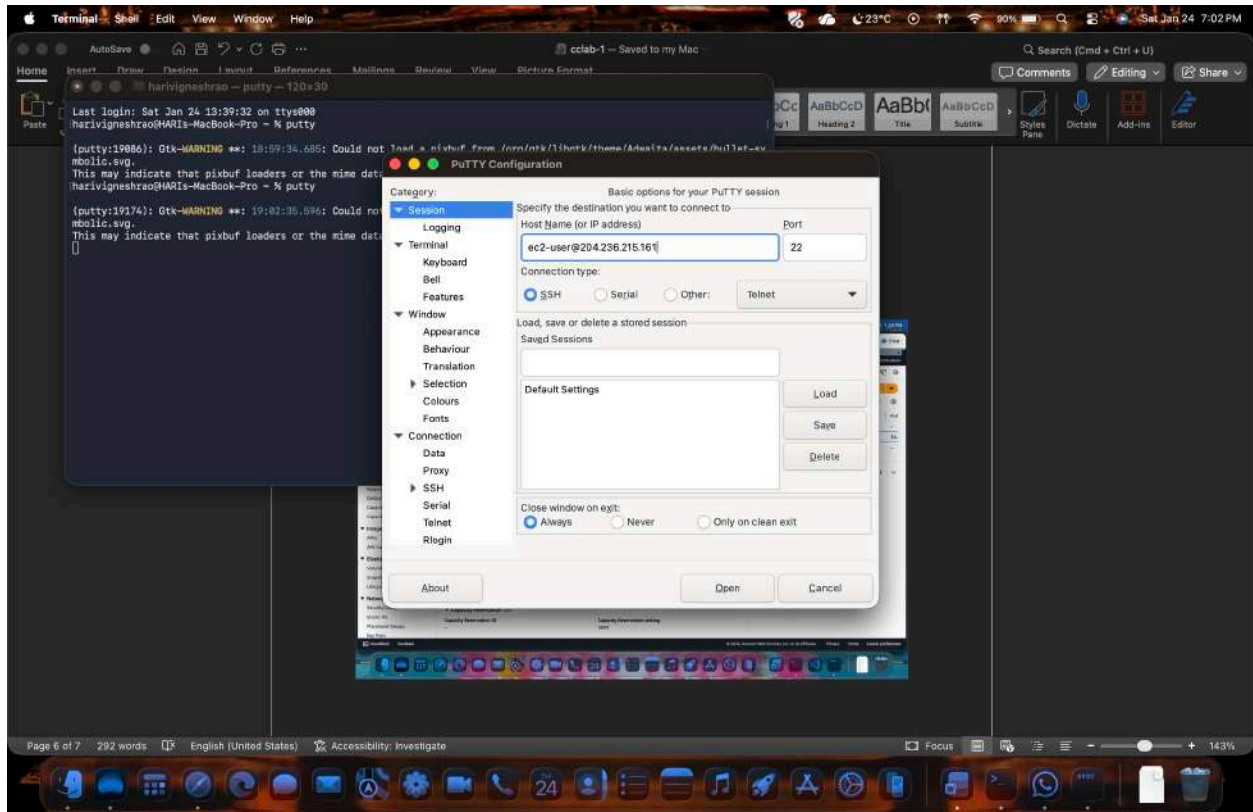


◆ STEP 2: Open PuTTY

1. Launch **PuTTY**
2. In **Session** screen:

Host Name (or IP address): EC2—CONNECT—COPY (DNS NAME)

ec2-user@<PUBLIC-IP>

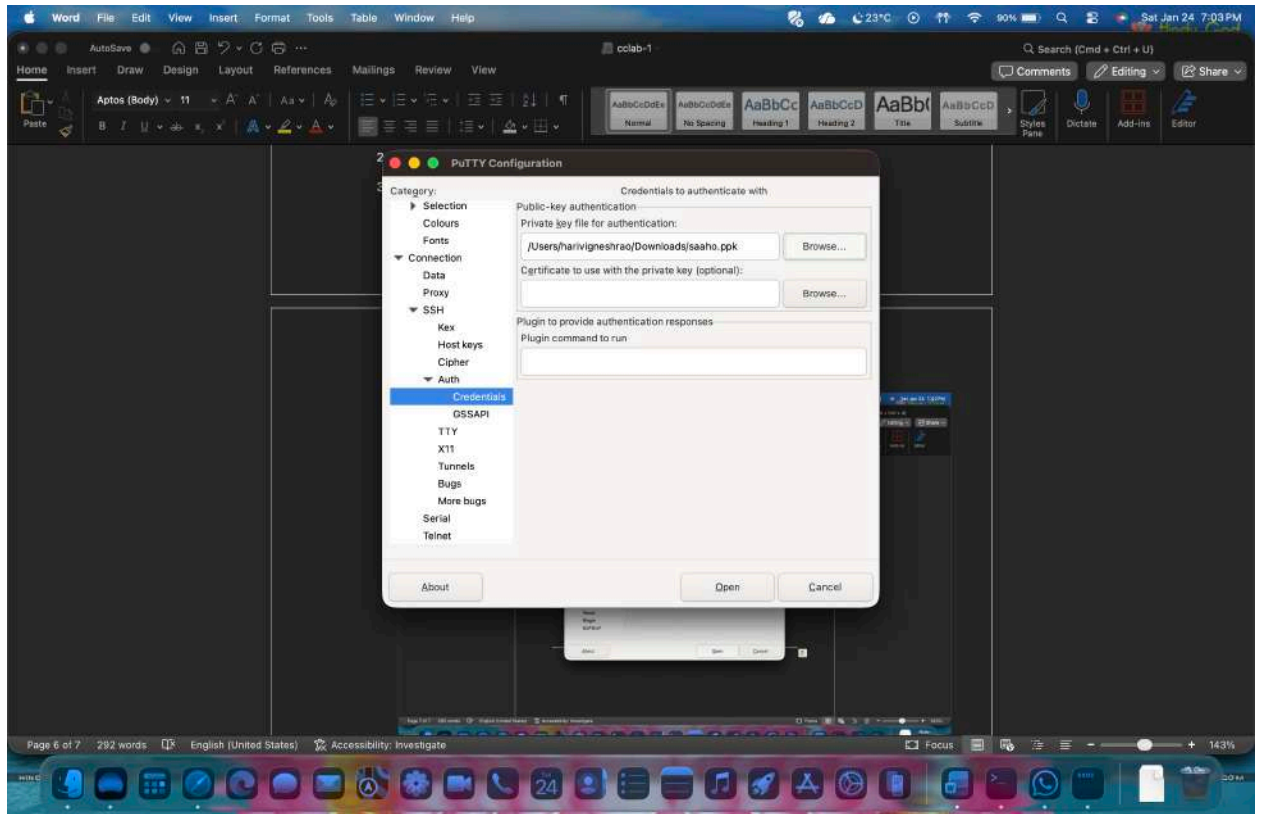


◆ STEP 3: Attach the .ppk Key

In PuTTY left menu, go to:

Connection → SSH → Auth → Credentials

1. Click **Browse**
2. Select your **.ppk file**
3. Click **Open**



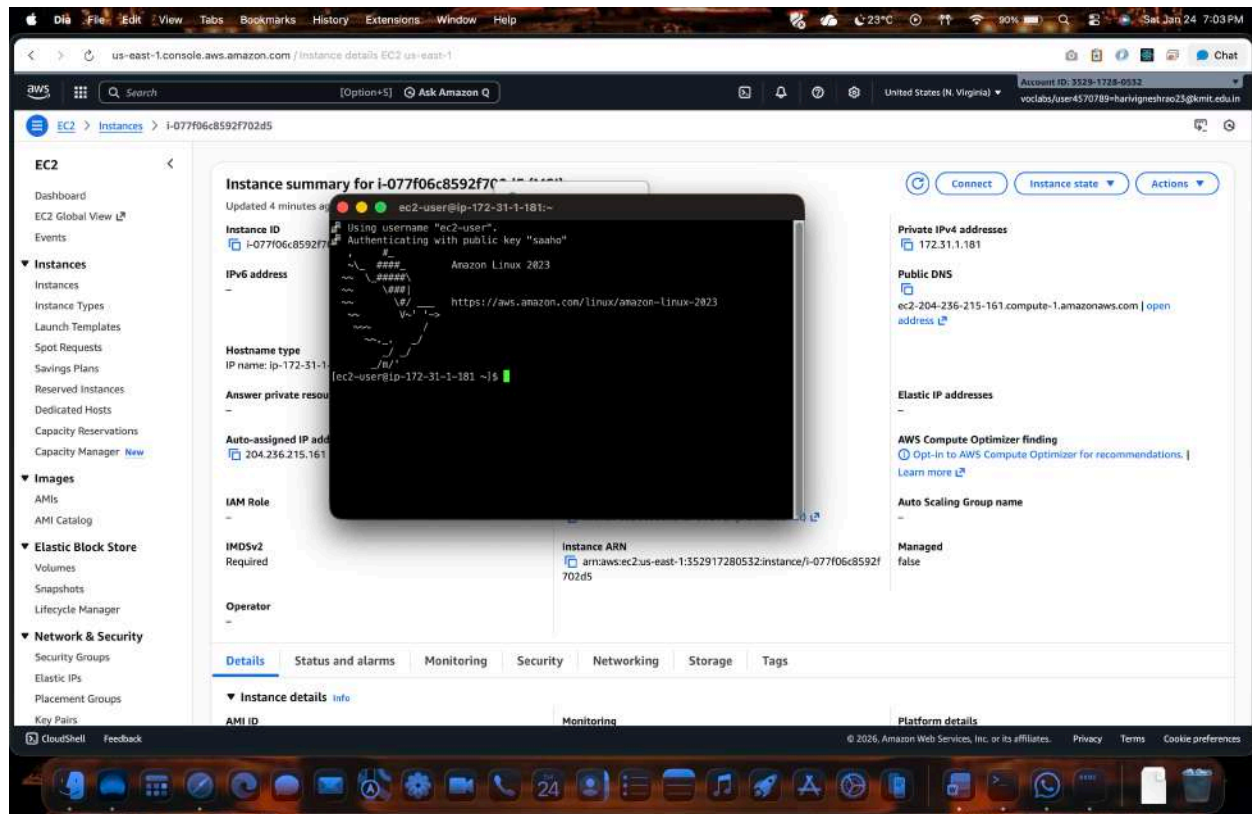
◆ STEP 4: Connect to EC2

Click **Open**

- First time → click **Yes** (security alert)
- PuTTY will log in automatically as:

ubuntu

🎉 **You are now connected to your Amazon EC2 instance**



Connect EC2 Ubuntu Instance Using PuTTY (Windows) ---**USING PEM KEYPAIR**

PREREQUISITES

You must have:

- ☒ **EC2 Ubuntu instance running**
- ☒ **Public IPv4 address (or Elastic IP)**
- ☒ **Key pair (.pem file) used while launching EC2**
- ☒ **PuTTY & PuTTYgen installed**

👉 Download PuTTY:

<https://www.putty.org>

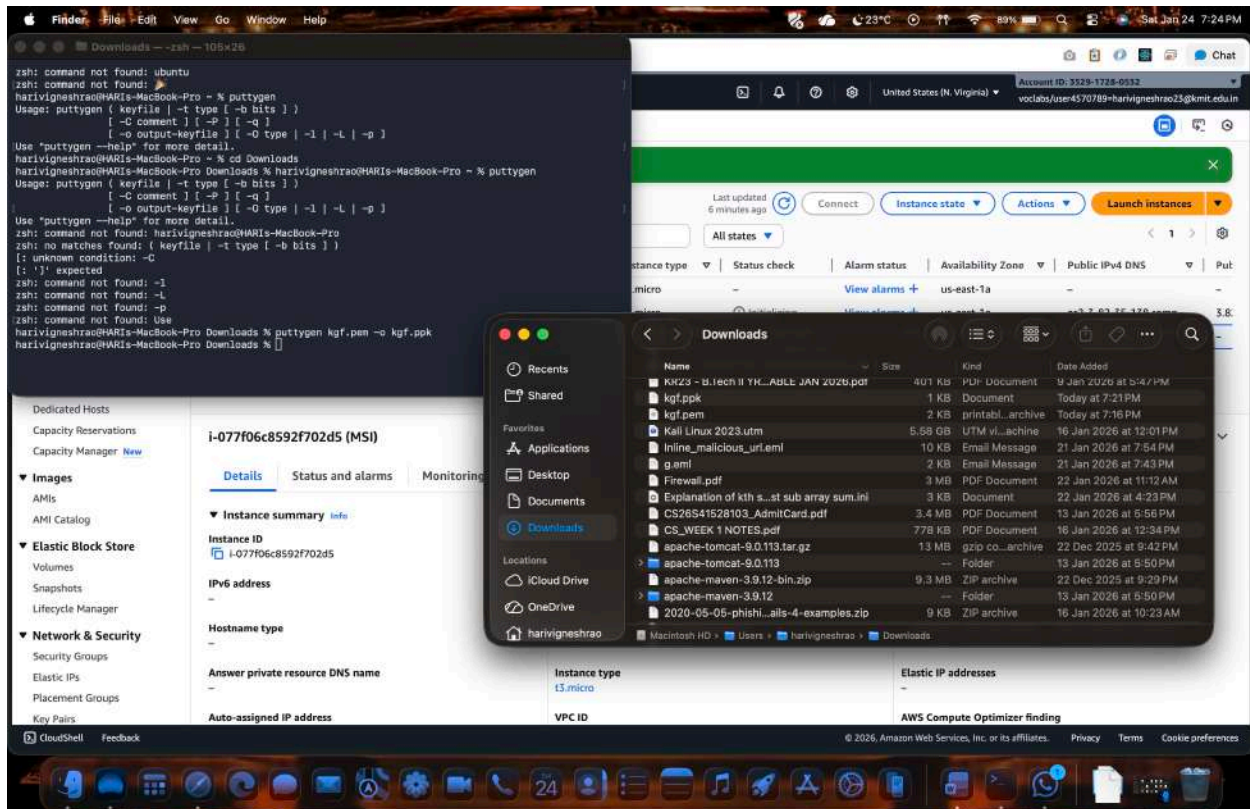
◆ STEP 1: Convert .pem to .ppk using PuTTYgen

PuTTY cannot use .pem directly.

Steps:

1. Open **PuTTYgen**
2. Click **Load**
3. Select your .pem file
(change file type to *All Files*)
4. Click **Open**
5. Click **Save private key**
6. Save as keyname.ppk

✓ Conversion done



◆ STEP 2: Get EC2 Public IP

Go to:

AWS Console → EC2 → Instances---CONNECT –COPY **DNS NAME**

Successfully initiated termination (deletion) of i-077f06c8592f702d5

Instances (1/3) info

Find instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Put
my_vs1	i-0B461ba0e11899457	Terminated	t3.micro	-	View alarms +	us-east-1a	-	-
MSI	i-0bad70e70a21d57e8	Running	t3.micro	Initializing	View alarms +	us-east-1a	ec2-3-82-35-138.comp...	3.8
MSI	i-077f06c8592f702d5	Terminated	t3.micro	-	View alarms +	us-east-1a	-	-

i-0bad70e70a21d57e8 (MSI)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary info

Instance ID i-0bad70e70a21d57e8

IPv6 address -

Hostname type IP name: ip-172-31-1-187.ec2.internal

Answer private resource DNS name -

Auto-assigned IP address -

Public IPv4 address 3.82.35.138 | open address

Instance state Running

Private IP DNS name (IPv4 only) ip-172-31-1-187.ec2.internal

Instance type t3.micro

Private IPv4 addresses 172.31.1.187

Public DNS ec2-3-82-35-138.compute-1.amazonaws.com | open address

Elastic IP addresses -

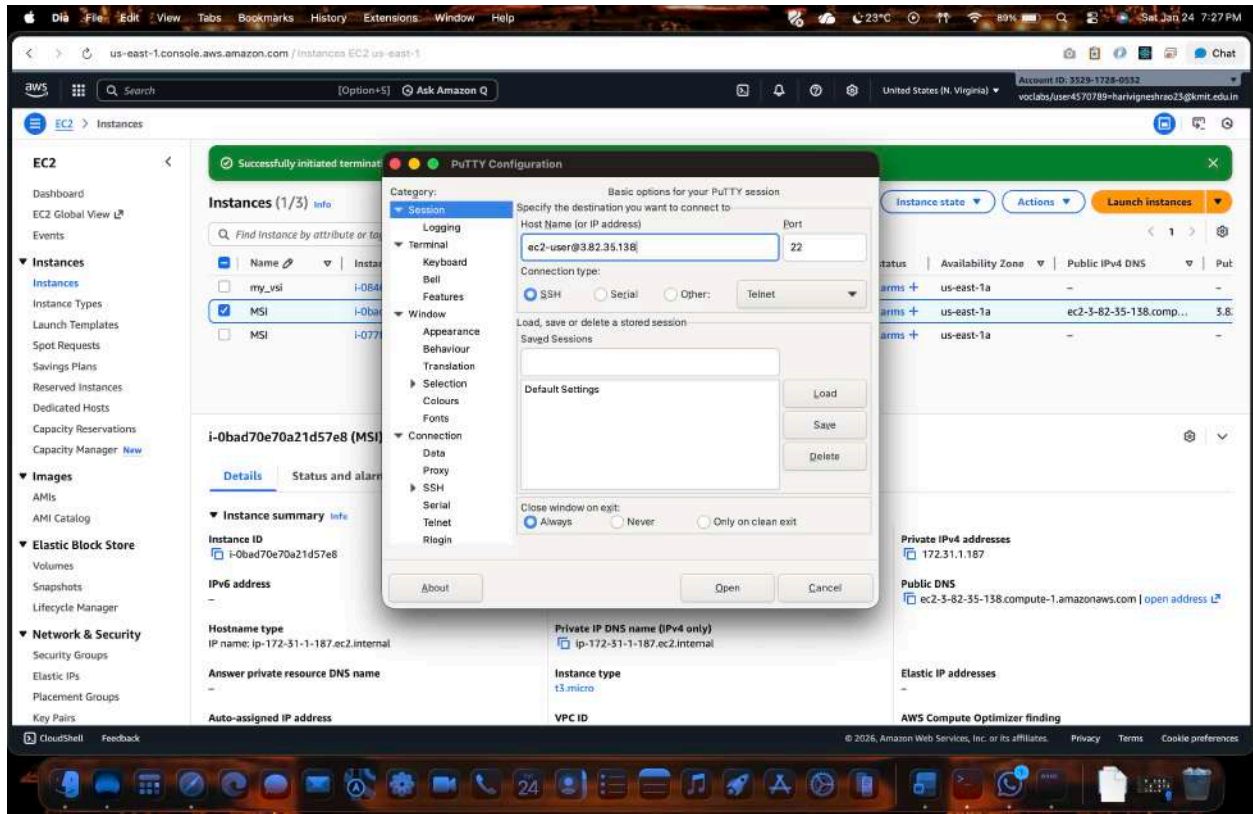
AWS Compute Optimizer finding

◆ STEP 3: Configure PuTTY

1 Open PuTTY

Session:

- Host Name (or IP address): **dns name**

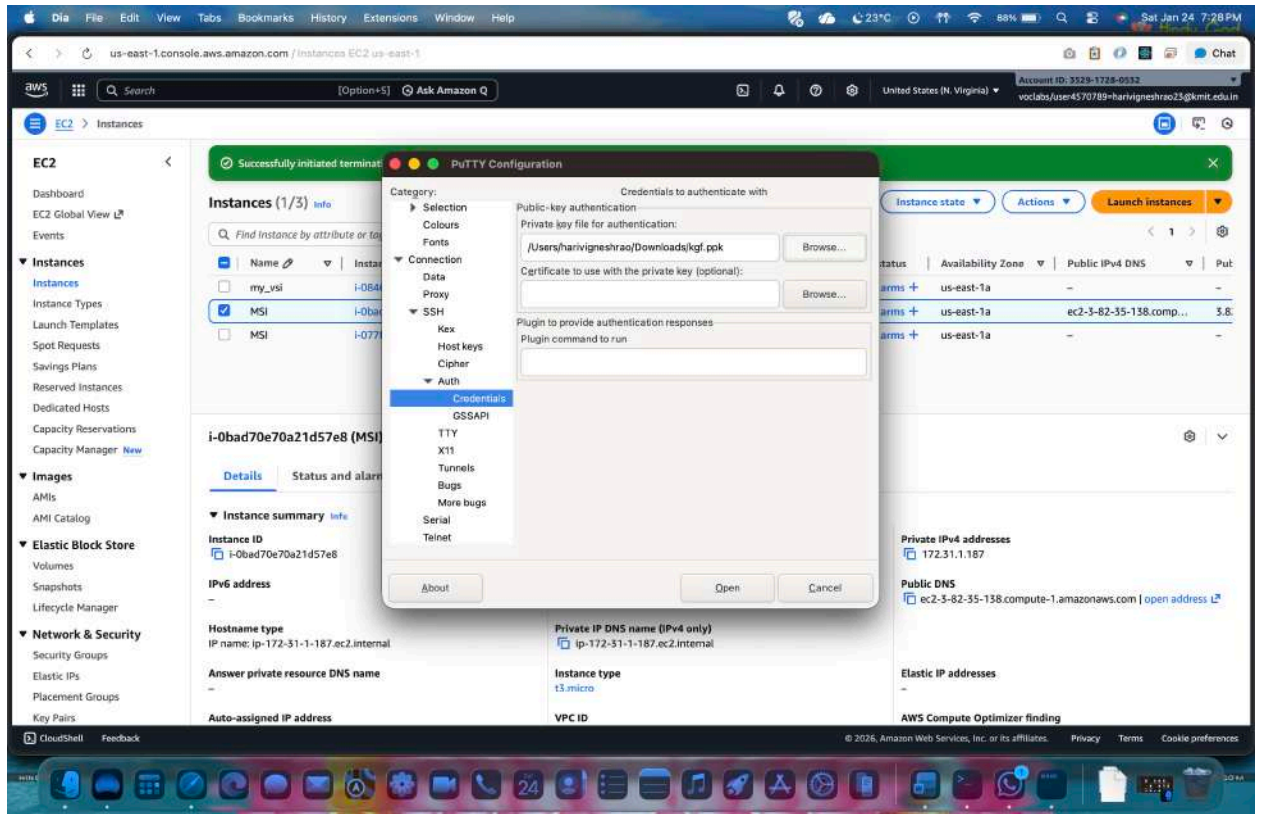


2 Load Private Key

Go to:

Connection → SSH → Auth → Credentials

- Click **Browse**
- Select your .ppk file



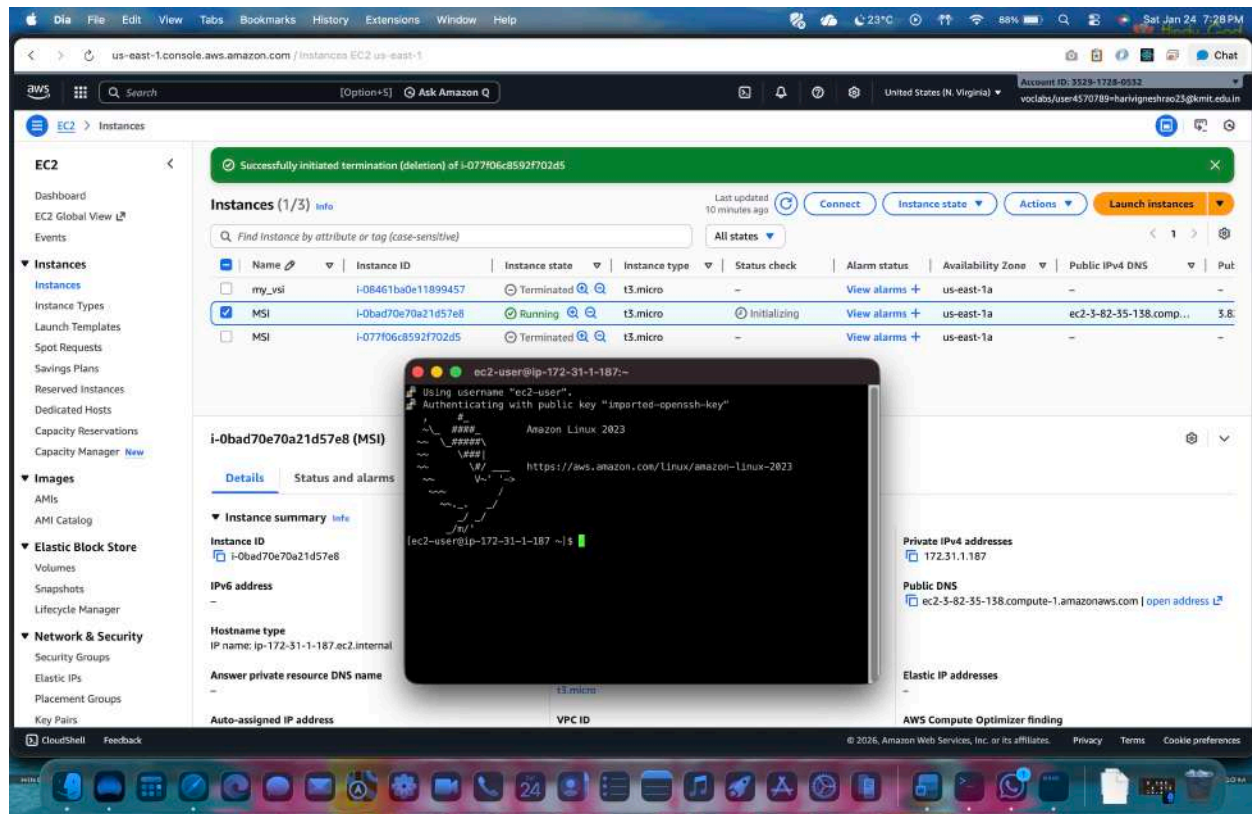
◆ STEP 4: Connect

Click Open

- First time → click **Yes** (security alert)
- Login happens automatically as:

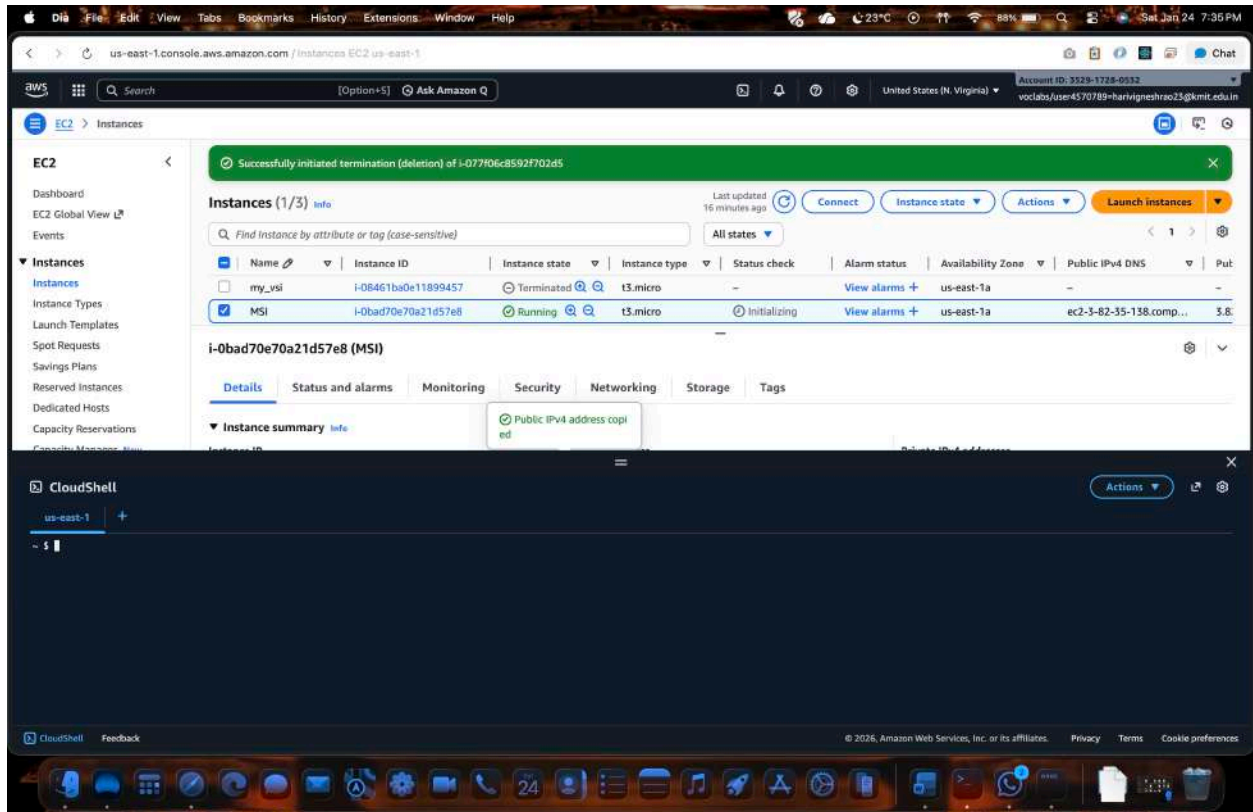
ubuntu

🎉 You are now connected to Amazon EC2

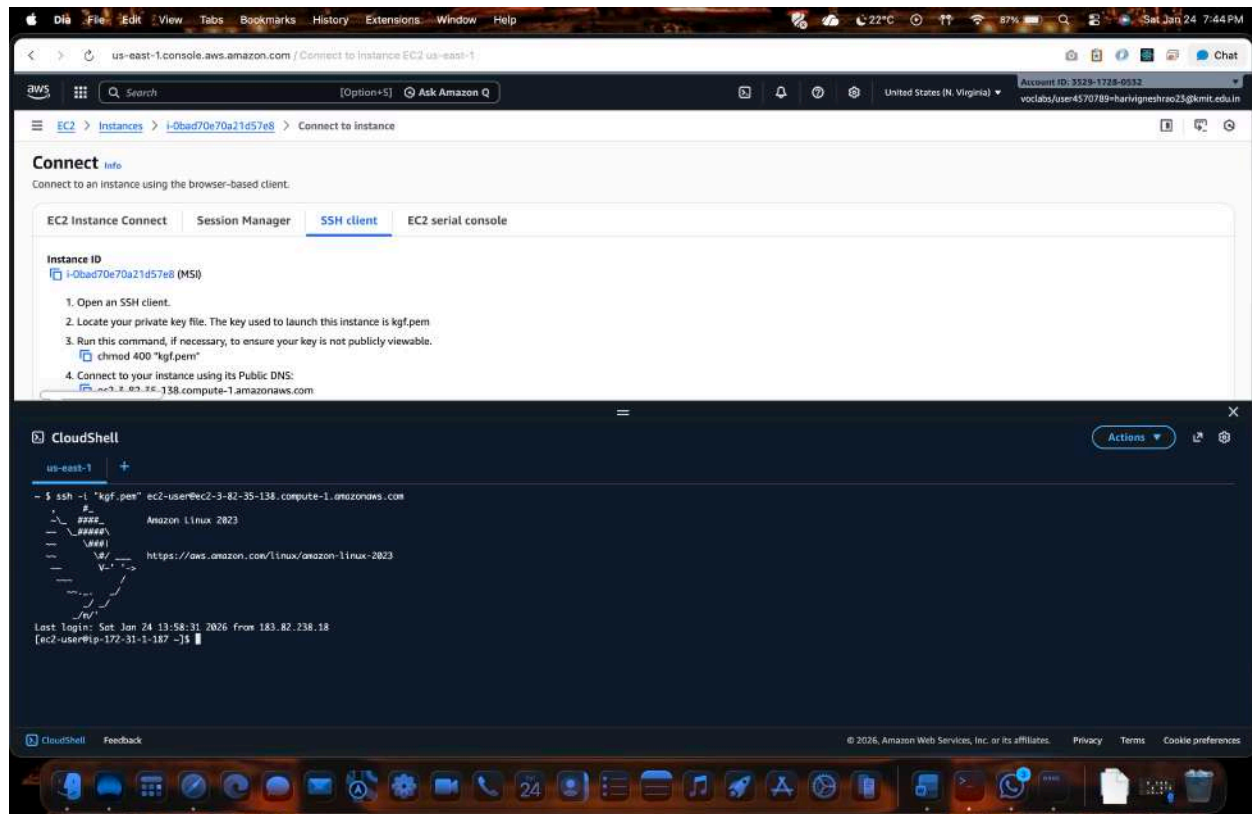


Connecting to EC2 Using AWS CloudShell

1. **Create the EC2 instance** using a key pair (.pem).
2. **Log in to AWS Console** and click on the **CloudShell** icon (terminal icon on the top-right).
3. Once CloudShell opens, go to **AWS Services** → **EC2**.



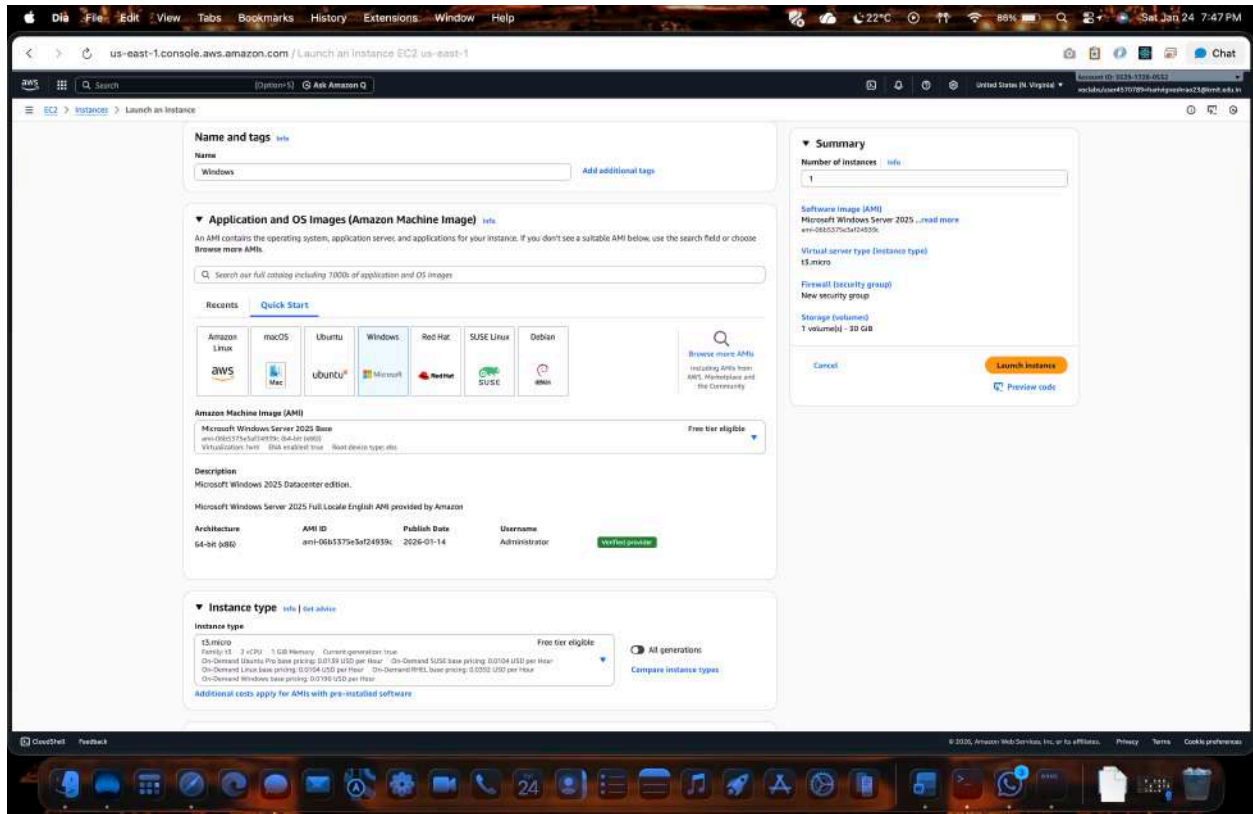
4. Select your **EC2 instance** and click **Connect**.
5. Choose the **SSH client** tab.
6. **Copy the ssh -i command** shown there.
7. **Paste the command in CloudShell** and try to run it.
 ✗ It will fail with an error like:
“key pair is not available / no such file or directory”
 (because the key file is not yet in CloudShell).
8. Now, in **CloudShell**, click on the **“+” (Actions)** icon on the right side.
9. Select **Upload file** and upload the **key pair (.pem)** from your local machine.
10. After upload, **copy the ssh -i command again** from **EC2 → Connect → SSH client**.
11. **Paste and execute the command** in CloudShell.
12. ✓ You are now **successfully connected to the EC2 instance**.



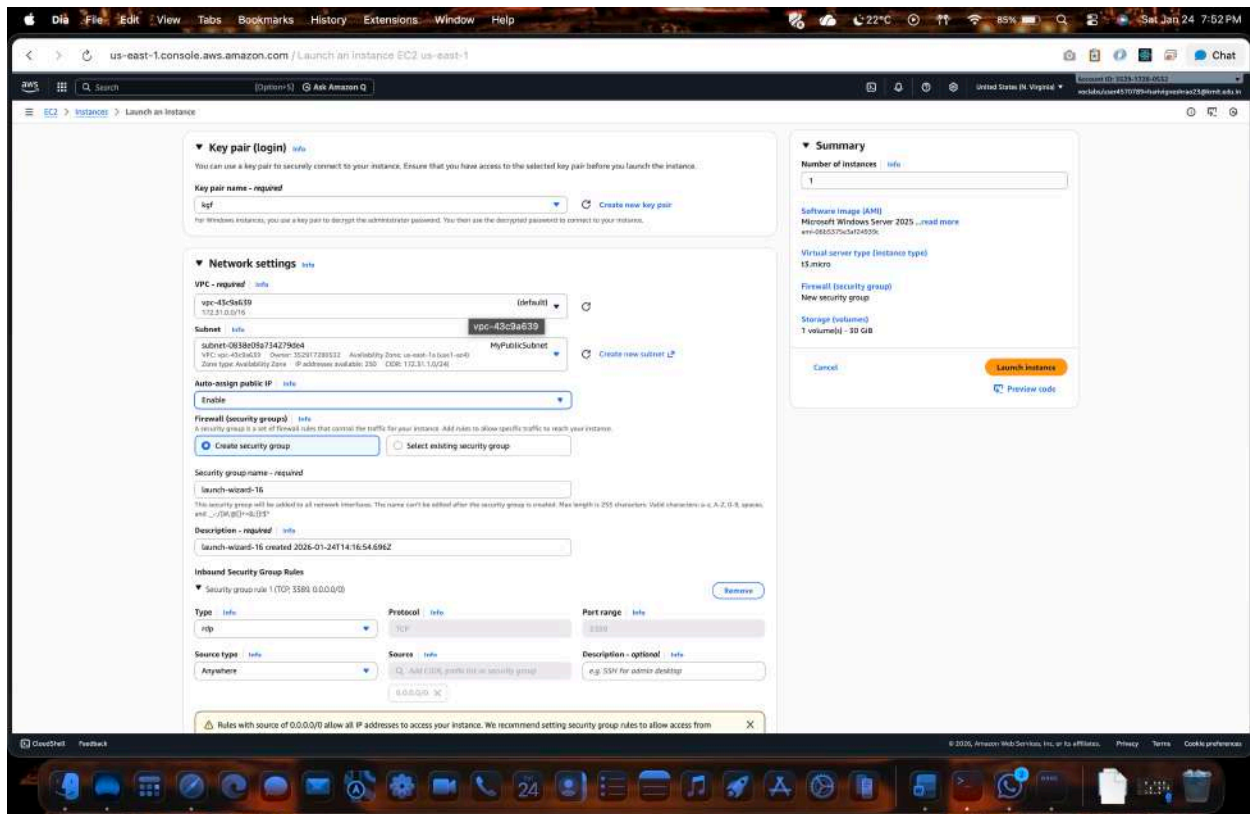
Step-by-Step: Create Windows EC2 Instance and Connect via RDP

Step 1: Launch a Windows EC2 Instance

1. Log in to your **AWS Management Console**.
2. Navigate to **EC2 → Instances → Launch Instances**.
3. Select an **Amazon Machine Image (AMI)**:
 - Choose a **Windows Server** AMI (e.g., Windows Server 2019 or 2022).
4. Choose an **Instance Type** (e.g., t2.micro for free tier or as per your requirement).

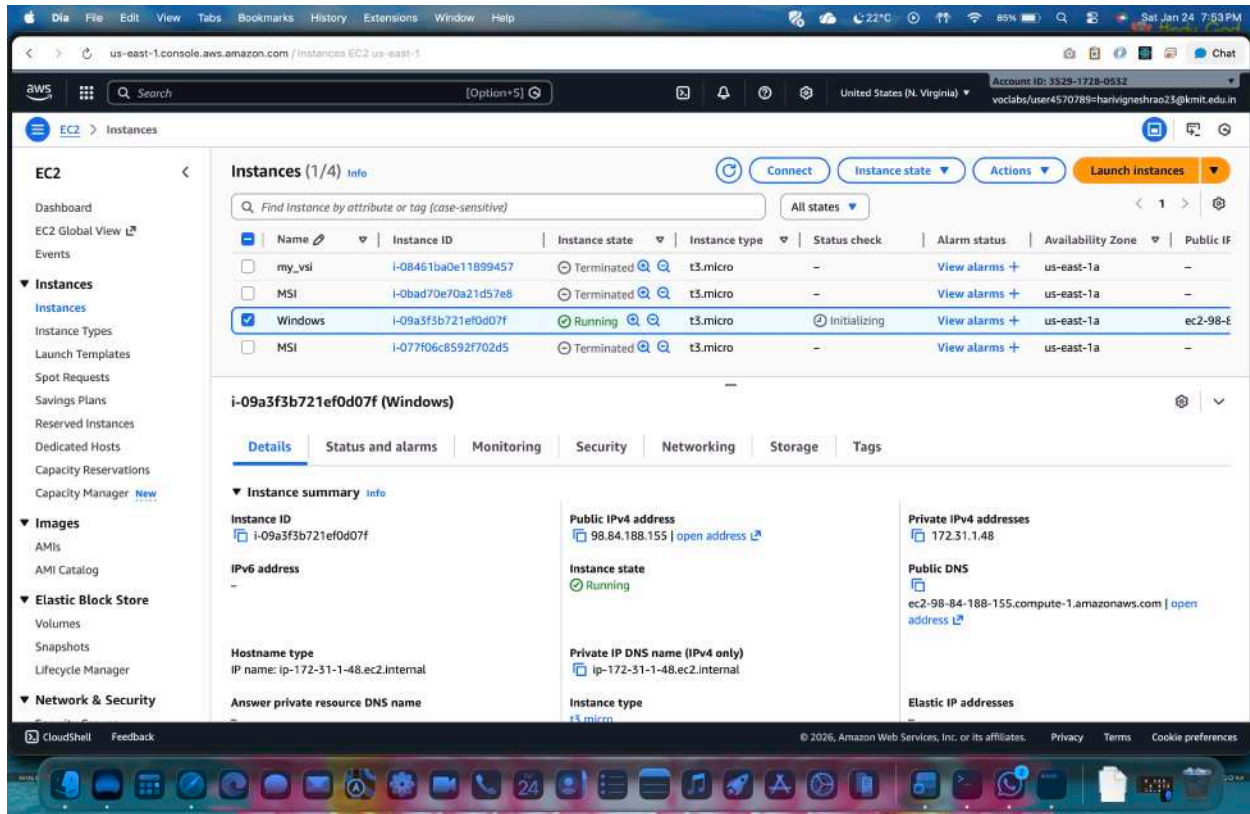


5. Configure **Instance Details** as needed.
6. Add **Storage** if required (default is usually sufficient).
7. Configure **Security Group**:
 - Ensure **RDP port 3389** is allowed.
 - Source can be **My IP** (for security) or a wider range if needed.
8. Select or **create a new Key Pair**:
 - Download the .pem file (keep it safe; it's needed to decrypt the password).
9. Click **Launch Instance**.



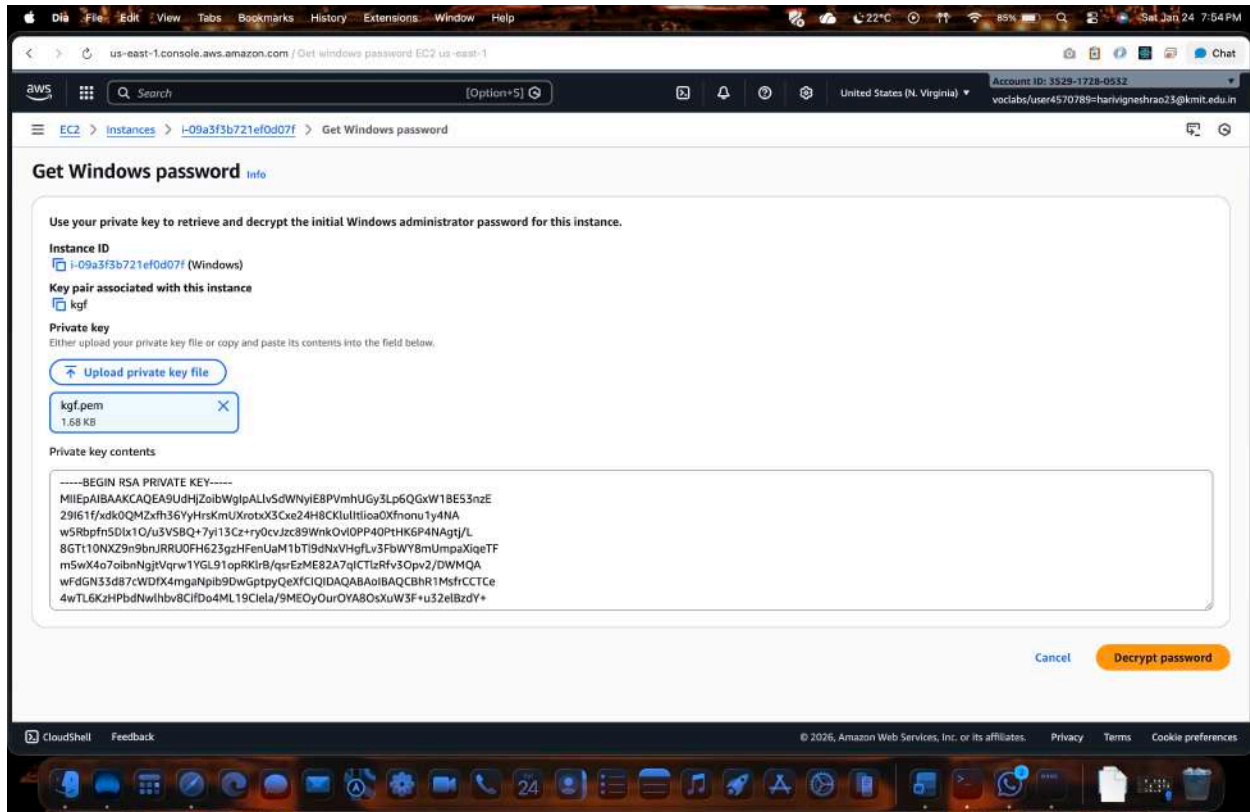
Step 2: Get the Public DNS / IP

1. After the instance is running, go to **EC2 → Instances**.
2. Select your instance and note the **Public DNS (IPv4)** or **Public IP**.



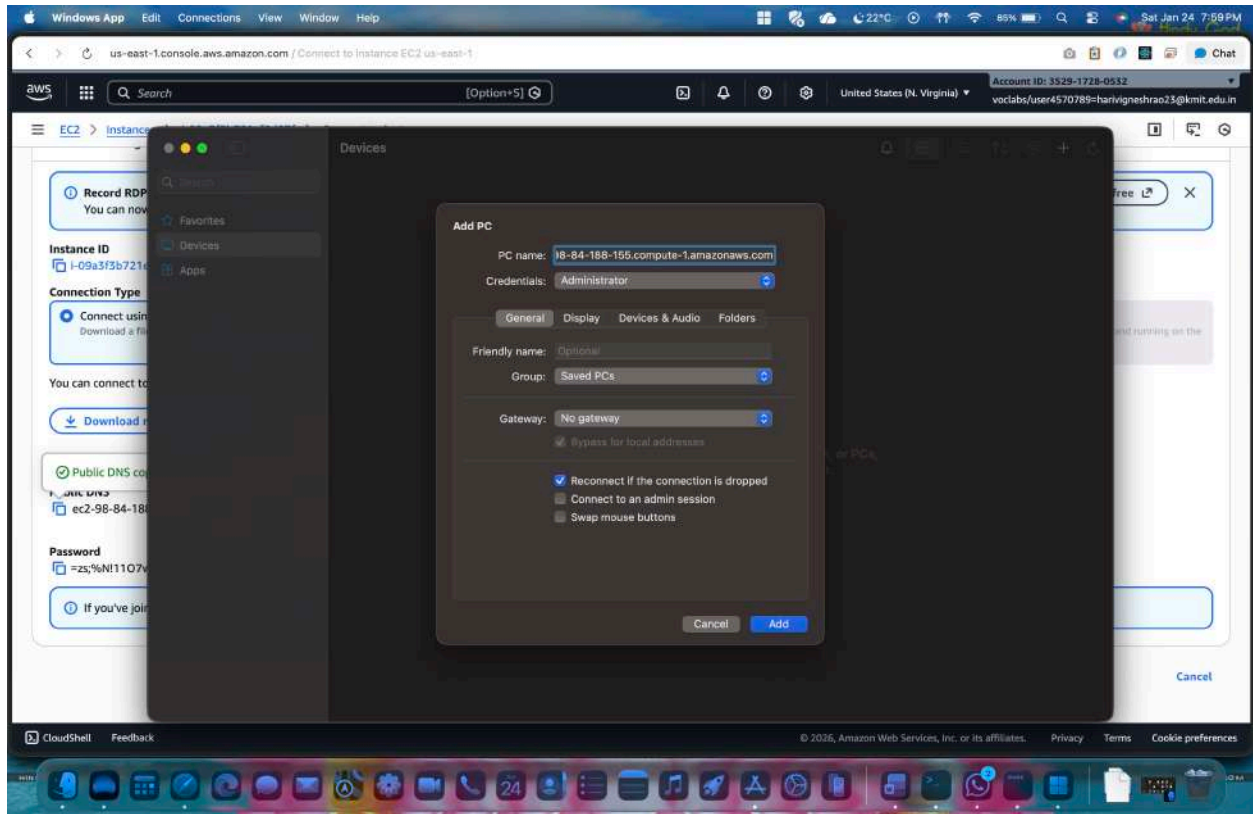
Step 3: Retrieve the Windows Administrator Password

1. Select your instance → Click **Connect** → **RDP Client**.
2. Click **Get Password**.
3. Upload your **Key Pair (.pem)** file.
4. Click **Decrypt Password**.
5. Copy the **Administrator password** shown.



Step 4: Connect via RDP

1. On your **local Windows machine**, open **Remote Desktop Connection** (mstsc).
2. In the **Computer** field, enter the **Public DNS** or **IP address** of the EC2 instance.
3. Click **Connect**.
4. In the login prompt:
 - **Username:** Administrator
 - **Password:** Paste the decrypted password from Step 3.
5. Click **OK**.



Step 5: You're connected

- Your Windows EC2 instance desktop should appear.
- You can now use it as a normal Windows machine.

